Ex No. 13 **App UI Design (Android – Kotlin)**

Aim

To design an interactive and user-friendly Android application UI using layouts, widgets, and views in Kotlin.

Procedure

1. Open Android Studio and create a new project (Empty Activity).
2. Set project name as AppUIDesign, language as Kotlin, and minimum SDK as API 21.
3. Open activity_main.xml file.
4. Use ConstraintLayout to design the UI.
5. Add UI components:
    - TextView (title display)
    - EditText (user input)
    - Button (click action)
6. Open MainActivity.kt.
7. Connect UI elements using findViewById.
8. Add click listener for the button.
9. Display Toast message and update TextView dynamically.
10. Run the application on emulator or physical device.

XML Code (activity_main.xml)

```
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/txtTitle"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Welcome"
      android:textSize="22sp"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintEnd_toEndOf="parent"/>

   <EditText
      android:id="@+id/edtName"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:hint="Enter Name"
      android:layout_margin="16dp"
      app:layout_constraintTop_toBottomOf="@id/txtTitle"
```

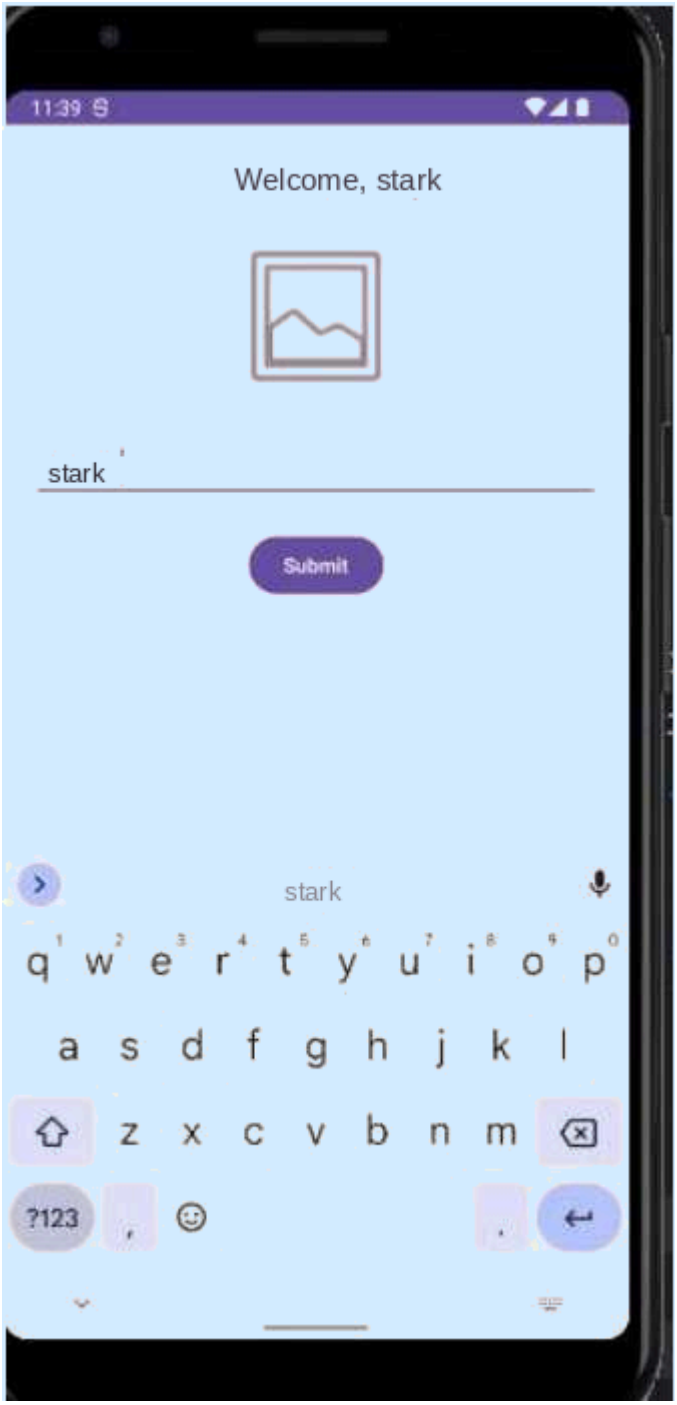
```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"
        app:layout_constraintTop_toBottomOf="@id/edtName"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

Kotlin Code (MainActivity.kt)

```
package com.example.appuidesign

import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val edtName = findViewById<EditText>(R.id.edtName)
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)
        val txtTitle = findViewById<TextView>(R.id.txtTitle)

        btnSubmit.setOnClickListener {
            val name = edtName.text.toString()

            if (name.isNotEmpty()) {
                txtTitle.text = "Welcome, $name"
                Toast.makeText(this, "Hello $name", Toast.LENGTH_SHORT).show()
            } else {
                Toast.makeText(this, "Enter name", Toast.LENGTH_SHORT).show()
            }
        }
    }
}
```

Result

The Android application UI was successfully designed using ConstraintLayout. The app takes user input through an EditText, and on button click, it displays a Toast message and updates the TextView dynamically based on the entered name.


11:39
Welcome, stark
stark
Submit
stark
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
?123